605.486 – Android Development

Module 22
RESTful Web Services

REST

- Representational State Transfer
- Data represented as “resources”
  - URIs describe individual resources or collections of resources
- Interaction using HTTP Methods
  - GET, POST, PUT, DELETE

REST URIs

- http://server:port/path/type
  - Collection of all resources of a type
- http://server:port/path/type/id
  - Specific instance of that type
- Parameters (unofficial)
  - Add at end after ?
  - http://server:port/api/person?name=Scott

REST Methods

- GET - read data from the server
- POST – create a new resource instance
- PUT – update an instance
- DELETE – delete an instance

- Instance data as JSON, XML, other…

Example 1 – REST Skeleton (server)

- Servlet
  - Can use frameworks like Jersey
  - Keeping it simple for this example

JSON

- JavaScript Object Notation
- Actual JavaScript code
- Describes objects
- Strict tree; no cycles
- http://json.org

JSON Object

```
{
    "name" : "Jenny",
    "phone" : "867-5309",
    "email" : "jenny@tutone.com"
}
```

JSON List

```
[
    {
        "name" : "Jenny",
        "friends" : [ {…}, {…} ],
        "email" : "jenny@tutone.com"
    }
]
```

Example 2 – Storing People (server)

- Person class
- Helper methods
- Simple code in servlet
- Test via REST Console Chrome extension

Example 3 –Setup (client)

- Person (can be different impl)
- Util class
- HttpMethod enum
- Result (code/string)
- ResultProcessor
- RequestTask (worker)

Example 4 – List People (client)

- List people
- MainActivity
- Layout
- Menu (for creating people later)

Example 5 – Create People (client)

- EditActivity
  - Layout
  - Menu
- MainActivity
  - startActivity – pass null id

Example 6 – Edit People (client)

- Click on person in list to edit
- startActivity passing id

Example 7 – Delete People (client)

- Contextual ActionBar
- Multiple Selection in list
- Run through selections and delete

Example 8 – Polling (client)

- Watch for changes on server
- Constantly ask “anything new?”
- Wasteful!!!
  - Bandwidth!!!
  - Battery!!!
  - Server resources!!!
- (We’ll fix this in GCM module)